

Decision Trees vs Deterministic Approaches for Detecting Phishing in Human and LLM-written Emails, and URLs

Group 6:

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June 2026

Abstract

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Phishing remains one of the most dominant threats to authentication systems, by relying on social engineering rather than technical exploitation to obtain sensitive information. The rapid growth of Large Language Models (LLMs) has made generating convincing phishing content even easier, while making detection increasingly more challenging. This study compares AI-based Decision Tree classifiers and deterministic rule-based approaches for detecting phishing in both emails and URLs, while also investigating how LLM-generated content can impact phishing detection.

Machine learning models were developed using TF-IDF vectorization and Decision Tree classification. The models include an LLM detector for emails, as well as separate phishing detectors for human-written and LLM-written emails, and a URL classifier. The performance of these models was compared against deterministic rule-based systems that were based on rules and heuristics. Experiments were conducted using an email dataset containing 4,000 human and LLM-written emails, and a URL dataset containing just over 650,000 URLs.

The AI approach achieved significantly higher accuracy than the deterministic approach. For email phishing detection, the AI model reached 98.38% accuracy compared to 50.38% for the deterministic method. For URL phishing detection, the AI model achieved 83.75% accuracy while the deterministic method achieved 12.75%. The LLM detector achieved 99.75% accuracy. Results further showed that all human-written phishing emails were correctly identified by the AI system, whereas 1.5% of LLM-generated phishing emails were not detected. In contrast, deterministic methods failed to detect any of the LLM-generated phishing emails.

The findings suggest that machine learning models are generally better at detecting modern

phishing attacks, particularly those generated by LLMs. Although deterministic approaches remain computationally inexpensive, they struggle with evolving phishing attacks, emphasizing the need for adaptive AI-based models.

1 Introduction

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Phishing is one of the most commonly used attacks against authentication systems. Instead of trying to break through technical vulnerabilities in security systems, attackers trick users into revealing their login credentials or other sensitive information. Phishing attacks most commonly take the form of deceptive emails and malicious URLs that mimic legitimate services to harvest credentials and personal data[1, 2].

The use of Large Language Models (LLMs) has made it even harder to detect phishing attacks. Previously, phishing emails were often identified through poor grammar, spelling mistakes, and unnatural language. However, with modern LLMs, highly convincing emails are easily generated with minimal effort containing more professional language, coherent structure, and sometimes personalized content [3, 4]. This has reduced many of the traditional indicators that users and automated deterministic systems often rely on to detect phishing attacks.

Existing research has mostly treated phishing detection and AI-generated text detection separately. Studies on phishing detection have primarily focused on identifying malicious emails and URLs using machine learning [1, 5, 6, 7, 8, 9, 10] and rule-based techniques [11], while research on AI-generated content focuses on distinguishing human-written content from AI-generated content, as well as focusing on both generating and detecting phishing content using LLMs [3]. Atawneh and Aljehani [8] demonstrated that deep learning models are effective

for email phishing detection, and reviews of future phishing detection techniques [12] have highlighted the rapid shift from rule-based to AI-driven approaches in the recent years.

This paper addresses this gap by comparing machine learning Decision Tree classifiers and deterministic rule-based approaches for phishing detection in both emails and URLs. Additionally, this study investigates whether identifying LLM-generated emails can improve the performance of phishing detection. Leading to evaluating the effect of LLM-generated content on phishing detection accuracy.

Our research focuses on addressing the following research questions:

- RQ1 - Which approach, between AI Decision Trees and Deterministic models, is more accurate in detecting phishing in URLs and human/LLM-written emails?
- RQ2 - What is the relationship between human and LLM phishing emails?
- RQ3 - What is the effect of LLM generated emails on phishing detection?

2 Methodology

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2.1 Datasets

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The AI models were trained on two distinct datasets, one for the email and LLM-detector models, and one for the URL model.

- Email Dataset - The email dataset contains four thousand emails in total, split in two sets of two thousand human-written and llm-written emails respectively. Each of those sets is split in two sets of one thousand legit and phishing emails [13].
- URL Dataset - The URL dataset contains 651,191 URLs, out of which 428,103 are benign or safe URLs, 96,457 defacement URLs, 94,111 phishing URLs, and 32,520 malware URLs [14].

2.2 Email Phishing Detection

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Reviewer: Jon Zdovc
Contributor: Jon Zdovc

Each email is first run through an LLM-Detector AI model to be classified as either human or LLM-written. Then, according to the classification, a human-written email phishing classifier AI model or a LLM-written email phishing classifier AI model is used to classify it as either a phishing or legit email.

The Deterministic models classify the email as either phishing or legit no matter what the LLM-Detector produces, because determining LLM-usage deterministically is out of the scope of this paper, since we only care about comparing the deterministic approach to the AI-based approach.

2.3 URL Phishing Detection

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A URL will be classified directly as either benign, phishing, defacement, or malware using both an AI model and a Deterministic one to determine the performance of each approach.

2.4 Data Collection

Main Writer: Shaked Gayer
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Results from all AI and Deterministic models for both emails and URLs are collected using data collection scripts for further statistical analysis to determine relationships and compare the performance between the two approaches.

3 Implementation

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The software implemented includes the AI and deterministic models, an API that makes requests to those models, a website and web extensions that use the API, and data collection scripts to perform experiments.

3.1 Software Versions and Libraries

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The AI models, data collection scripts, and API were written in Python, while the website and web extensions were built using HTML, CSS,

and JavaScript. The software versions and used libraries are shown in Table 1.

Software / Library	Version
Python	3.12
Flask	3.0.1
Flask-Cors	5.0.0
Joblib	1.3.1
Pandas	3.0.1
Scikit-Learn	1.8.0
Python-WHOIS	0.9.6

Table 1: Software versions

3.2 AI Models

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All AI models are built on sklearn’s implementation of Decision Trees (DTs). They first convert all emails and URLs in their respective dataset to TF-IDF (term frequencyinverse document frequency) vectors. The vocabulary is derived from the dataset ($\leq 10^4$ terms), with 12-grams and English stop-word removal for email classification, and 13-grams without stop-word filtering for URL classification. This TF-IDF and n-gram approach to URL classification is consistent with the NLP-based feature extraction strategy shown to be effective by Sahingoz [6].

Afterwards, a pipeline architecture is used to build the DTs. They are then calibrated using sigmoid calibration with 5-fold cross-validation to offset the overconfidence that DTs usually produce and to provide deeper insight into the accuracy of the model. Next, they are trained using an 80/20 training/test set split. Finally, the accuracy is used to evaluate the model’s performance. Using the above-described architecture, the following models were built:

- LLM-Detector - Trained on the whole email dataset. Built to classify an email as LLM or human-written.
- Human-written Email Phishing Detector - Trained only on human-written legit and phishing emails. Built to classify a human-written email as legit or phishing.
- LLM-written Email Phishing Detector - Trained only on LLM-written legit and phishing emails. Built to classify a LLM-written email as legit or phishing.
- URL Phishing Detector - Trained on the URL dataset. Built to classify a URL as benign, phishing, defacement or malware.

3.3 Deterministic Models

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The deterministic models use manual rules and heuristics instead of machine learning. Unlike the AI models, the deterministic models do not learn patterns from data and instead rely on predefined indicators that are commonly found in phishing attacks [11]. Each deterministic model assigns risk scores based on specific indicators and use threshold-based decision making to conclude how likely an email or URL is to be a phishing attack.

3.3.1 Email Deterministic Model

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The deterministic email phishing detector evaluates both the email subject and body using four heuristic categories, targeting linguistic and structural patterns widely associated with phishing emails [1, 3]:

- **Urgent language** (+20 points) – Detects phrases commonly used to pressure recipients into taking immediate action, such as *urgent*, *immediate action*, *account suspended*, *verify your account*, *security alert*, and *final notice*.
- **Generic greetings** (+15 points) – Detects non-personalized greetings such as *Dear Customer*, *Dear User*, and *Dear Member*, which are frequently used in mass phishing attacks.
- **Requests for sensitive information** (+30 points) – Detects references to credentials and personal information including passwords, social security numbers, credit card information, and bank account details.
- **Suspicious links** (+20 points) – Detects URLs containing raw IP addresses or common URL shortening services such as Bitly, TinyURL, and Ow.ly.

The risk score is then calculated by summing the points associated with all the triggered heuristics and capping the result at 100. Emails that receive a score of 50 or larger are classified as phishing, and emails receiving a score below 50 are classified as safe.

3.3.2 URL Deterministic Model

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The deterministic URL phishing detector is based on a weighted scoring system based on research of features that were identified in previously detected phishing websites [6, 5, 11, 15]. Each URL is analyzed using multiple independent checks which are executed in parallel to improve the model’s performance.

The first checker compares the URL against the OpenPhish[16] blacklist feed, which checks if it been previously marked as phishing. If a match is found, the URL is immediately classified as malicious by giving it the maximum risk score of 100.

For URLs that are not found in the blacklist, the following heuristic checks are performed:

- **Regex-based phishing patterns** – Detects characteristics frequently associated with phishing URLs, including the “@” symbol, suspicious redirections (“//” appearing after the protocol section), hyphens in domain names, extra domain nesting, unusually long URLs, URL shortening services, and misleading use of the term “https” within the domain name.
- **IP address detection** – Identifies URLs that use IPv4 addresses or hexadecimal-encoded IP addresses instead of domain names.
- **Domain age analysis** – Uses WHOIS records to determine the age of the domain. Domains younger than six months receive the highest risk score, while older domains receive progressively lower scores.
- **HTTPS verification** – Checks whether the URL uses HTTPS encryption. URLs that do not use HTTPS are considered higher risk.
- **Suspicious top-level domains (TLDs)** – Detects TLDs commonly known with phishing attacks, such as *.tk*, *.ml*, *.ga*, *.xyz*, *.top*, and *.click*.

The final risk score is calculated through a weighted combination of the individual checker outputs. HTTPS status accounts for 30% of the final score, regex-based features accounts for 25%, IP-address detection accounts for 20%, domain-age analysis accounts for 15%, and TLD analysis accounts for 10%. The resulting score is then categorized into one of three URL classifications:

- Benign (0–40)
- Defacement (41–70)
- Malware (71–100)

The deterministic models provide clear and easy to interpret classifications on whether a URL is considered to be suspicious or not, due to the specific characteristics that contributed to the final classification.

3.4 API

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The API is built using Python. Each AI model is saved after training as a joblib file and called in the API. The Deterministic models are directly called. It integrates requests to each model and returns their results as JSON objects with the predicted label and phishing probability through GET and POST methods. The API serves as the backbone of the website, web extensions, and data collection scripts, as they all depend on it for results.

3.5 Data Collection Scripts

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Two separate data collection scripts were implemented to facilitate the experiment processes.

3.5.1 Email Data Collection Script

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The email data collection script extracts all instances from the email dataset with the option of a custom even split between human and LLM-written both legit and phishing emails. It makes calls to the API to every AI Email model as well as the Deterministic Email and LLM Detector models on every single email instance. Finally, it stores the returned classifications in a CSV table, sorted by the method used (AI or Deterministic) for further analysis.

3.5.2 URL Data Collection Script

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Similar to the Email Data Collection Script, the URL Data Collection Script extracts all URL instances from the URL dataset and makes calls to the API to the AI and Deterministic URL Phishing Classifier. It stores the results in a

CSV table, sorted by the method used (AI or Deterministic) for further analysis.

3.6 Website & Web Extensions

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3.6.1 Website

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A static website was built using HTML, CSS, and JavaScript where users can interact through a Graphical User Interface (GUI) with the models by providing them new instances of URLs and emails for classification.

The website’s JavaScript makes calls to the API in the following manner:

- Emails - Upon input of an email subject and body from the user, an initial API call to the LLM-detector is made. After the classification is returned as either human or LLM-written, a subsequent API call to either the Human-written/LLM-written Email Phishing Detector or the Deterministic Email Phishing Detector (according to the user’s stated preference) is made to determine the phishing probability. The result is returned as a numerical value and displayed on the GUI, as well as the result from the LLM-Detector, also as a numerical value.
- URLs - Upon input of a URL from the user, an API call is made to either the AI URL Phishing Detector or the Deterministic URL Phishing Detector, based on the user’s stated preference. The result from either of the models is returned as a numerical value and displayed on the GUI to the user.

3.6.2 Web Extensions

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Web extensions were developed for the purpose of users interacting with the models in a more convenient manner. Both web extensions require a Chromium-based browser to be used and both use the Chrome Extension platform Manifest V3 for communicating with the browser. The Email Web Extension only works on Gmail.

- Email Web Extension - It makes calls to Manifest V3 to acquire the DOM objects that store the subject and body of an email on Gmail. It then makes calls to the model API to get the phishing and LLM-usage scores according to our architecture and displays an HTML GUI object to the user.
- URL Web Extension - A background Chrome worker makes calls to our AI URL Phishing Detector through the API when a new connection request is detected by the browser. The URL is extracted using Manifest V3. Then, if the URL is classified as phishing, defacement or malware, the browser blocks the connection and a HTML GUI object is displayed on the webpage stating the URL is malicious.

4 Experiments

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The performance of all models was collected using the data collection scripts that make API calls to each model for every instance in the email and URL datasets respectively and store those results in CSV tables for further analysis. The columns of those tables are as outlined in Tables 3 and 2.

For RQ1, the returned classifications were compared to the original labels for both emails and URLs in order to determine the accuracy of each model. Then, the accuracies of the AI and Deterministic models were compared to determine the better performing approach.

For RQ2 and RQ3, the likelihood of correctly classifying an LLM-written phishing email using the AI and Deterministic models was analyzed in order to determine if an LLM-written email is more likely to be classified as phishing or not. The false-negative (phishing but classified as legit), and false-positive (legit but classified as phishing) rates were calculated to determine the effect of LLM-written emails on phishing detection.

5 Results

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5.1 AI Models

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Column name	Description
URL	The URL extracted from the URL dataset
original_label	The label of the URL from the dataset -benign, defacement, phishing, malware
method	The prediction method used - AI or Deterministic
prediction	The predicted label from the model

Table 2: URL Data Fields

Column name	Description
email_content	The email extracted from the email dataset
original_label	The label of the email from the dataset -legit or phishing
type	The type of the email - Human/LLM-written
method	The prediction method used - AI or Deterministic
prediction	The predicted label from the model
llm_detection	The predicted label from the LLM Detector - AI/LLM-written or Human-written

Table 3: Email Data Fields

Model	Accuracy	Precision (Legit, Phishing)	Recall (Legit, Phishing)	F1-Score (Legit, Phishing)
Human-written Email Phishing Detector	0.98	0.98, 0.98	0.97, 0.98	0.98, 0.98
LLM-written Email Phishing Detector	0.93	0.93, 0.93	0.93, 0.93	0.93, 0.93

Table 4: AI Email Models Results

Model	Accuracy	Precision (Human, LLM)	Recall (Human, LLM)	F1-Score (Human, LLM)
LLM Detector	0.98	0.98, 0.98	0.98, 0.98	0.98, 0.98

Table 5: LLM Detector Model Results

Model	Accuracy	Precision (Benign, Defacement, Malware, Phishing)	Recall (Benign, Defacement, Malware, Phishing)	F1-Score (Benign, Defacement, Malware, Phishing)
URL Phishing Detector	0.98	0.91, 0.93, 0.91, 0.81	0.98, 0.93, 0.76, 0.57	0.94, 0.93, 0.83, 0.67

Table 6: AI URL Model Results

5.2 Data Analysis

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5.2.1 Email Data

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The accuracy of each method (AI or Deterministic), calculated by comparing the "original_label" and "prediction" columns in the Email data table gave the following results, as shown in Figure 1:

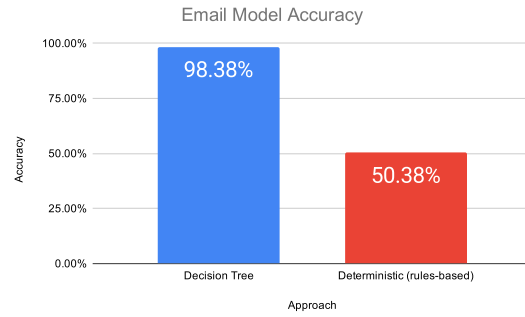


Figure 1: Email Model Accuracy per type

The LLM Detector accuracy, calculated by comparing the "type" and "llm_detection" columns in the data produced an accuracy of **99.75%**.

False Positive Rate: Calculated by comparing the accuracy between the "original_label" and "prediction" columns (how many legit emails were flagged as phishing) for each method and email type. For the AI method, both Human-written and LLM-written legit emails were incorrectly flagged as phishing at a rate of **2.5%**. The deterministic method had a **0%** false positive rate for both.

True Positive Rate: Calculated by comparing the accuracy between the "original_label" and "prediction" columns (how many phishing emails were correctly classified) for each method and email type. For the AI method, **100%** of human-written phishing emails were correctly identified, whereas **98.5%** of LLM-written phishing emails were caught. In other words, **1.5%** of LLM-written phishing emails escaped detection.

The deterministic classification method failed to catch any of the LLM-written phishing emails with **0%** detection, while it caught a total of **1.5%** of human-written ones.

5.2.2 URL Data

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The accuracy of each method (AI or Deterministic), calculated by comparing the "original_label" and "prediction" columns in the URL data table gave the following results, as shown in Figure 2:

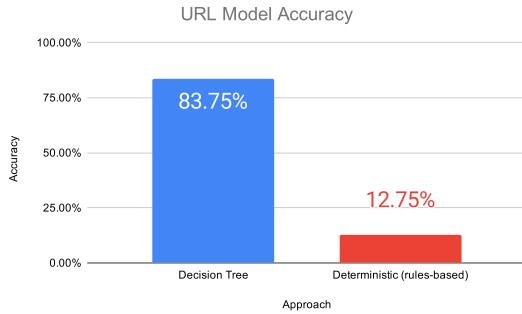


Figure 2: URL Model Accuracy per type

6 Discussion

Main Writer: Olivier van Leeuwen

Reviewer: Theodotos Neokleous

Contributor: Theodotos Neokleous

The results clearly show that the AI-based approach significantly outperformed the deterministic rule-based approach for detecting phishing attacks in both emails and URLs. When classifying emails, the AI-based model achieved an accuracy of 98.38%, while the deterministic rule-based model achieved an accuracy of 50.38%. When classifying URLs, the AI-based model achieved an accuracy of 83.75%, while the deterministic rule-based model achieved only 12.75% accuracy. These findings directly answer our first research question (RQ1) by showing that Decision Tree-based machine learning models provide significantly better accuracy when it comes to detecting phishing attacks compared to rule-based deterministic models.

This gap in performance is explained by the ability of machine learning models to learn from the training data [17, 18]. The Decision Tree classifiers learn statistical relationships between patterns and phishing behavior that might not be obvious. On the other hand, deterministic

models rely on predefined indicators that can easily be bypassed by attackers when they know what to avoid. Modern phishing attacks tend to avoid the traditional phishing checks such as spelling mistakes, suspicious formatting, or obvious malicious keywords. This reduces the accuracy of rule-based deterministic approaches as they constantly need updates and can only cover a certain number of checks.

Our findings indicate that there is a difference between human-written and LLM-generated phishing emails [3], which relates to our second research question. The human-written email phishing detector achieved an accuracy of 98%, while the LLM-written email phishing detector achieved an accuracy of 93%. Additionally, as shown by the experiment scripts, the AI-based model detected 100% of the human-written phishing emails, but only 98.5% of the LLM-generated phishing emails. Although the difference is quite small, it still does suggest that LLM-generated phishing content includes characteristics that make it slightly harder to detect phishing attacks [4].

The deterministic rule-based model detected only 1.5% of the human-written phishing emails and was unable to detect any of the LLM-generated phishing emails. This suggests that many of the traditional phishing attack indicators are not present in LLM-generated content [3]. These findings address the third research question and highlight the impact of LLM-generated content on phishing detection. LLMs can produce professionally written coherent content with correct grammar, which rule-based deterministic models rely on to detect phishing attacks, causing them to be less effective.

These findings are consistent with recent literature suggesting that AI-generated phishing attacks are becoming increasingly difficult to detect. Heiding et al. demonstrated that large language models can generate phishing emails that are comparable in quality to those written by humans, reducing many of the linguistic indicators traditionally used to identify phishing attempts [3]. Additionally, evaluations involving human participants showed that AI-generated phishing attacks can achieve high levels of effectiveness as they are becoming even harder to distinguish from human-written emails [4]. These observations support the results of this study, where LLM-generated phishing emails were slightly harder to detect than human-written phishing emails and significantly reduced the effectiveness of deterministic rule-based approaches.

There are several limitations to consider when interpreting the results of these findings. Firstly, the email dataset contains only 4,000 entries, which may not fully represent all the different types of real-world phishing attacks. Sec-

only, the dataset may not contain the most recent phishing techniques and therefore it is possible that the models may not detect attacks with evolved strategies. Thirdly, the experiments were performed using static datasets rather than live datasets that are updated with current phishing attacks. Lastly, the deterministic rule-based models rely solely on a set number of manually selected rules and thresholds. A larger number of rules or alternative configurations could potentially increase the performance of the model's ability to detect phishing attacks.

Future work could include investigating different machine learning architectures to compare the differences between them. Further testing with carefully handcrafted phishing attacks could help provide further insight into the best way to design the rules for the deterministic models to improve their performance accuracy. Additionally, incorporating real-time phishing attack datasets could improve the performance of the machine learning models as the datasets get continuously updated with evolving attacks.

7 Conclusion

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This study investigated the effect of AI-based Decision Tree classifiers versus deterministic rule-based approaches for detecting phishing attacks in emails and URLs, with particular emphasis on the impact of LLM-generated content in phishing emails.

Answering the first research question, "Which approach, between AI Decision Trees and Deterministic models, is more accurate in detecting phishing in URLs and human/LLM-written emails?", the results gathered showed that AI-based models significantly outperformed the deterministic models in both detecting phishing emails and URLs. The AI-based system achieved 98.38% accuracy for email phishing detection and 83.75% accuracy for URL phishing detection, while the deterministic rule-based system achieved an accuracy of 50.38% for emails and 12.75% for URLs.

Answering the second research question, "What is the relationship between human and LLM phishing emails?", the results gathered confirmed there is indeed a relationship between the performance of phishing detection and the content of the email. LLM-generated phishing emails proved to be slightly more difficult to detect compared to human-written phishing emails, indicating that LLM-generated content can mimic legitimate emails more closely than human-written ones.

Answering the third research question, "What is the effect of LLM generated emails on phishing detection?", the results of the study found that LLM-generated emails substantially reduced the effectiveness of deterministic rule-based phishing detection models. The deterministic approach failed to successfully identify any of the LLM-generated phishing emails, compared to the AI-based approach. This highlights the growing threat caused by constantly evolving LLM-generated phishing attacks.

Overall, the findings of this study suggest that traditional deterministic rule-based approaches are no longer enough to defend against modern phishing attacks on their own. Machine learning models provide improved adaptability and phishing detection performance. As LLM-generated phishing attacks become increasingly more common, AI-based detection models play a crucial role in building defenses against phishing attacks.

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